



1st International Conference on Soft Computing Application in Smart Grid and Renewable Energy (SAGAR2027)

26th & 27th February, 2027

Department of Electrical & Computer Engineering
&

Department of Electrical Engineering

Ajay Binay Institute Of Technology (Autonomous), Cuttack



About SAGAR2027 :

The 1st International Conference on Soft Computing Application in Smart Grid and Renewable Energy (SAGAR2027) will be held on February 26-27, 2027, at ABIT(Autonomous) in hybrid mode. The conference is jointly organized by the Department of Electrical and Computer Engineering and Department of Electrical Engineering.

SAGAR2027 focuses on the application of soft computing techniques which includes artificial intelligence, machine learning, neural networks, fuzzy logic, evolutionary algorithms, optimization, data science, and analytics that address challenges in smart grids and renewable energy systems.

The conference aims to bring together researchers, academicians, industry professionals, and students to discuss recent advances in intelligent technologies for power systems, renewable energy integration, energy management, forecasting, control, optimization, and reliability enhancement. It provides a platform for knowledge sharing, collaboration, and exploration of future trends in sustainable and smart energy systems.

About Ajay Binay Institute of Technology (ABIT) Autonomous college :

Ajay Binay Institute of Technology (ABIT) is approved by AICTE, affiliated with BPUT, and has been granted Autonomous Status by UGC. ABIT group operates an Engineering College and an Industrial Training Institute in Cuttack, Odisha.

The group was established in 1993 with the aim to improve the availability of skilled technical manpower for industries as well as to enhance reach of technical education to rural Odisha.

The Institute is approved by AICTE, New Delhi, affiliated to Biju Patnaik University of Technology, Rourkela and accredited by NAAC. ABIT has a balanced distribution of technical streams with equal emphasis on the core sector and IT sector.

Important Date :

- Submission Open: July 6, 2026
- Submission Deadline: Nov 20, 2026
- Acceptance Notification: Dec 27, 2026
- Author Registration: Jan 1, 2027
- Camera Ready Paper Submission: Jan 15, 2027
- Conference dates: February 26-27, 2027

Registration fee

Category	Indian Author	Foreign Author
Full Registration	6000 INR	150 USD
Participation Fee	2000 INR	50 USD

*Note:

- Registration is valid for UG/PG students and PhD scholar with valid Id card, Academician and Industry professionals.
- At least one author registration is mandatory for each accepted paper.
- Co-authors who wish to attend the conference in offline mode must register under the Attendee category.

*Accepted and presented papers will be considered for publication in Springer Proceedings (subject to Springer approval). The proceedings are expected to be indexed in major databases such as Scopus.

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Venue :

Department of Electrical and Computer Engineering (ECE)
and Department of Electrical Engineering,
AJAY BINAY INSTITUTE OF TECHNOLOGY (AUTONOMOUS),
Markat Nagar, Cuttack Development Authority, Sector - 1,
Cuttack - 753014

SAGAR2027 Website



<https://sagar2027.vercel.app/>

For Paper Submission



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TECHNICAL TRACKS:

Track 1: Power Systems and Smart Grids

- Power generation, transmission, and distribution
- Power system operation and control
- Smart grid technologies and applications
- Renewable energy integration and grid integration challenges
- Grid resilience and reliability enhancement
- Energy storage systems and management
- Demand-side management and demand response strategies
- Smart Grid Planning and Operation

Track 2: AI & ML application and Cyber security

- AI-Based Intrusion Detection Systems (IDS)
- Deep Learning Applications
- Phishing and Spam Detection Using Machine Learning
- Malware Classification with Deep Learning
- Network Anomaly Detection Using AI
- Adversarial Attacks and Defenses in Machine Learning

Track 3: Application of optimization techniques

- power system optimization
- Vehicle Routing and Logistics Optimization
- Hyperparameter Optimization in Machine Learning
- Smart Grid Energy Management Optimization
- Production and Inventory Scheduling Optimization
- Resource Allocation in Cloud Computing Using Optimization Algorithms

Track 4: Power Electronics and Intelligent Control Systems

- Intelligent control of power converters
- Advanced inverter technologies
- Control strategies for renewable energy systems
- FACTS devices and smart grid applications
- Adaptive and robust control techniques
- Embedded systems for smart energy applications
- Automation in power systems
- Electric Vehicle and Charging station

Track 5: Emerging Technologies and Future Trends

- Internet of Things (IoT) and communication techniques
- Quantum computing applications in energy systems
- Green hydrogen technologies
- AI-driven carbon neutrality solutions
- Internet of Energy (IoE)
- Smart energy markets and trading
- Digital transformation in power systems
- Future challenges in renewable energy and smart grids

Paper Submission Link: <https://cmt3.research.microsoft.com/SAGAR2027>

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Tourist Attractions:

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|--------------------------------|--------------------------------|----------------------------|
| • Barabati Stadium | • Barabati Fort | • Chilika Lake |
| • Silvercity Boat Club | • Odisha State Maritime Museum | • Lingaraj Temple |
| • Shree Jagannath Temple, Puri | • Netaji Birth Place Museum | • Dhauli santi stupa |
| • Konark Sun Temple | • Cuttack Chandi Temple | • Khandagiri and Udayagiri |
| | | • Puri Sea Beach |



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